

Quiz: Lecture 1: Introduction to Cloud Computing

Q1. According to the lecture, what were the primary drivers for the evolutionary changes in distributed and cloud computing over the past 30 years?

- (A) A decline in the cost of personal computers
- (B) The invention of the internet
- (C) Applications with variable workloads and large data sets
- (D) Increased security protocols for data transmission

Q2. The lecture mentions that distributed computing uses multiple computers to solve large-scale problems, making it 'data intensive and network centric'. What concept does the lecture identify as the key distinguisher that makes modern cloud computing different from traditional distributed systems?

- (A) The use of message passing for communication
- (B) The virtualization of resources to create a flexible pool
- (C) The connection of computers over the internet
- (D) The goal of solving large-scale problems

Q3. Which of the following is NOT listed in the lecture as a form of high-throughput computing system that preceded or is related to clouds?

- (A) Computational grids
- (B) Peer to peer networks
- (C) Mainframe timesharing systems
- (D) Service oriented architecture

Q4. A startup needs to quickly deploy a server for a 3-month project. According to the lecture's example, what is the primary advantage of using a cloud service like AWS over purchasing their own server?

- (A) Guaranteed higher security for their data
- (B) Saving significant time in server setup
- (C) Access to more powerful CPU cores
- (D) Better integration with their existing private hardware

Q5. The lecture describes a 'pay as you go' model as a key feature of cloud computing. What is the formal term for this feature?

- (A) Massive Scale
- (B) On-demand Access
- (C) Data-intensive Nature
- (D) Utility Computing

Q6. Which major cloud provider is mentioned as offering the Elastic Compute Cloud (EC2) and Simple Storage Service (S3)?

- (A) Microsoft Azure
- (B) Google Compute Engine
- (C) Amazon Web Service
- (D) Rightscale

Q7. A developer wants to build and deploy an application using a specific programming framework provided by a cloud vendor, without needing to manage the underlying operating system or virtual machines. Which service model would be most appropriate?

- (A) Infrastructure as a Service (IaaS)
- (B) Software as a Service (SaaS)
- (C) Hardware as a Service (HaaS)
- (D) Platform as a Service (PaaS)

Q8. What is the primary difference between a public cloud and a private cloud as described in the lecture?

- (A) Public clouds use virtualization, while private clouds use bare-metal servers.
- (B) Public clouds are open to any paying customer, while private clouds are for a single company's internal use.
- (C) Public clouds are for storage only, while private clouds are for computation.
- (D) Public clouds are managed by the government, while private clouds are managed by corporations.

Q9. The lecture discusses the historical evolution of computing paradigms, quoting slogans from different eras. Who is credited with the slogan, 'the data center is the computer'?

- (A) John Gage of Sun Microsystems
- (B) David Patterson of UC Berkeley
- (C) Rajkumar Buyya of Melbourne University
- (D) Fernando Corbato of MIT

Q10. When describing the internal structure of a single-site cloud (datacenter), the lecture mentions a 'top of the rack switch'. What is the function of this component?

- (A) To provide power to all servers in the rack
- (B) To connect all the servers within a single rack
- (C) To connect the entire datacenter to the internet
- (D) To cool the servers located at the top of the rack

Q11. The lecture describes a shift from 'computation-intensive' to 'data-intensive' computing. In data-intensive computing, which resource metric becomes more important than CPU utilization?

- (A) The number of CPU cores
- (B) The speed of the processor clock
- (C) The I/O performance of disk and network
- (D) The amount of RAM per machine

Q12. Which of the following is presented as an open-source version of Google's MapReduce paradigm?

- (A) Cassandra
- (B) Hadoop
- (C) MongoDB
- (D) Sawzall

Q13. According to the lecture's economic breakeven analysis, under what condition is it more cost-effective for a company to build its own private cloud rather than use a public cloud?

- (A) When the project duration is very short (e.g., less than 6 months)
- (B) When the company requires a small number of servers
- (C) When the project duration is long (e.g., more than 6-12 months)
- (D) When the company prioritizes operational expenditure over capital expenditure

Q14. What is the concept of 'utility computing,' as envisioned by Fernando Corbato and described in the lecture?

- (A) A business model where software is sold with a perpetual license.
- (B) Computing resources should be as easily accessible as electricity or water from a utility company.
- (C) All computing should be done on personal, utility-sized devices.
- (D) A system where utility companies manage all internet traffic.

Q15. The lecture discusses the massive scale of cloud operations. Which application was mentioned as running on 180,000 servers by 2012?

- (A) Microsoft Bing
- (B) Yahoo Search
- (C) Google's Indexing Service
- (D) Facebook

Q16. What is the primary function of Amazon's EC2 service, as described in the lecture?

- (A) To provide scalable object storage for datasets
- (B) To provide raw, bare-metal hardware access
- (C) To provide compute services where users can run operating system images
- (D) To provide a database service for key-value stores

Q17. The lecture identifies four key features of today's clouds. Which of the following is NOT one of those four features?

- (A) Massive Scale
- (B) On-demand Access
- (C) Guaranteed Security
- (D) New Cloud Programming Paradigms

Q18. What metric does the lecture introduce to measure the power efficiency of a datacenter, defined as 'total facility power divided by IT equipment power'?

- (A) Water Usage Effectiveness (WUE)
- (B) Power Usage Effectiveness (PUE)
- (C) Server Utilization Ratio (SUR)
- (D) Energy Star Rating (ESR)

Q19. According to the lecture, what historical computing system is considered a 'precursor to the current cloud computing systems'?

- (A) ENIAC and early datacenters
- (B) Timesharing systems with terminals
- (C) Peer-to-peer (P2P) systems
- (D) The first personal computers (PCs)

Q20. What does Gilder's Law, as mentioned in the lecture, predict?

- (A) Processor speeds double every 18 months.
- (B) Network bandwidth doubles each year.
- (C) Storage capacity doubles every 12 months.
- (D) The value of a network is proportional to the square of its users.

Q21. A financial analytics company needs to process terabytes of transaction logs. The data is too large to move efficiently. What computing approach, highlighted in the lecture, is best suited for this scenario?

- (A) Computation-intensive computing on a supercomputer
- (B) Data-intensive computing where computation moves to the data
- (C) A private cloud focused on high CPU utilization
- (D) A SaaS application for financial modeling

Q22. Which of the following is presented as an example of Software as a Service (SaaS)?

- (A) Amazon EC2
- (B) Google App Engine
- (C) Google Docs
- (D) OpenStack

Q23. Why did the lecture suggest that Hardware as a Service (HaaS) is 'not a good idea'?

- (A) It is too expensive for most customers.
- (B) It is less flexible than Infrastructure as a Service.
- (C) It presents security risks.
- (D) It has been made obsolete by personal computers.

Q24. The lecture describes a '3-tier architecture' for a datacenter. What component acts as the 'frontend' in this architecture?

- (A) The core switches connecting the racks
- (B) The servers used for submitting jobs and receiving client requests
- (C) The storage nodes with SSDs
- (D) The individual compute nodes within a rack

Q25. According to the lecture, which technology 'makes it possible' for the growth of internet clouds as a new computing paradigm?

- (A) Message Passing Interface (MPI)
- (B) The development of the personal computer
- (C) Advances in virtualization
- (D) The invention of the relational database

Q26. What is the pricing model for Amazon S3 (Simple Storage Service) as described in the lecture?

- (A) Per CPU hour used
- (B) A one-time flat fee for unlimited storage
- (C) Based on the amount of space used, in GB per month
- (D) Based on the number of files stored, regardless of size

Q27. The lecture states that using online services can reduce operational costs. Why is this the case?

- (A) Cloud providers offer free services to gain market share.
- (B) The operational cost is eliminated because customers only pay for what they use.
- (C) Cloud services use more energy-efficient hardware.
- (D) The software licenses for cloud services are cheaper.

Q28. What is the key-value store system mentioned in the lecture that is noted for being significantly faster than MySQL?

- (A) Hadoop
- (B) MongoDB
- (C) Pig
- (D) Cassandra

Q29. The lecture mentions several cloud providers besides the 'big three' (Amazon, Microsoft, Google). Which of the following was listed as another provider?

- (A) Sun Microsystems
- (B) IBM
- (C) Salesforce
- (D) Intel

Q30. What is the primary characteristic of 'geographically distributed clouds'?

- (A) They consist of a single, massive datacenter in one location.
- (B) They are comprised of multiple datacenters in different locations connected by a network.
- (C) They rely exclusively on peer-to-peer networking instead of datacenters.
- (D) They are a theoretical concept that has not yet been implemented.

Q31. The historical overview of computing shows a cyclical pattern. The lecture suggests that modern datacenters are a repeat of which earlier era, but with different notions of computing?

- (A) The era of personal computers (PCs) in the 1980s.
- (B) The era of large datacenters like ENIAC in the 1940s-1960s.
- (C) The era of peer-to-peer (P2P) systems.
- (D) The era of timesharing systems in the 1970s.

Q32. According to Moore's Law as cited in the lecture, how often do processor capabilities double?

- (A) Every 9 months
- (B) Every 12 months
- (C) Every 18 months
- (D) Every 24 months

Q33. A company is building a web service that needs to serve millions of users. The lecture suggests a paradigm shift in software development for this scenario. What is this shift?

- (A) Developing software to be distributed and run on individual PCs.
- (B) Focusing on single-threaded performance.
- (C) Developing software to be delivered as a service to millions of users.
- (D) Writing software that requires minimal hardware resources.

Q34. What is the primary function of a 'core switch' in the datacenter architecture described?

- (A) To connect individual servers within a single rack.
- (B) To connect all the 'top of the rack' switches together.
- (C) To provide a frontend for job submission.
- (D) To manage power distribution to the entire datacenter.

Q35. Which company, according to the lecture, uses a chain of 24 MapReduce jobs for its indexing process?

- (A) Facebook
- (B) Yahoo
- (C) Amazon
- (D) Google

Q36. The lecture claims that 'the cloud is the computer'. Who is credited with this statement?

- (A) David Patterson
- (B) John Gage
- (C) Rajkumar Buyya
- (D) Fernando Corbato

Q37. For a problem to be classified as a 'cloud computing problem', the lecture suggests it should have one or more key features. Which scenario best exemplifies the 'data-intensive' feature?

- (A) Running a complex mathematical simulation that takes weeks on a single CPU.
- (B) A startup needing a web server for a few months.
- (C) Processing and analyzing petabytes of daily weblogs from a popular website.
- (D) Using an online document editor instead of a desktop application.

Q38. The lecture mentions that virtualization has enabled Infrastructure as a Service (IaaS) to subsume a different, less popular service model. Which model did IaaS subsume?

- (A) Platform as a Service (PaaS)
- (B) Software as a Service (SaaS)
- (C) Hardware as a Service (HaaS)
- (D) Database as a Service (DBaaS)

Q39. What is the primary role of storage backend nodes within a datacenter rack, as described in the lecture?

- (A) To run the main application logic.
- (B) To handle network routing between racks.
- (C) To be primarily used for storage purposes, often containing SSDs.
- (D) To act as a frontend for client requests.

Q40. Which of the following is NOT a service provided by Amazon Web Services mentioned in the lecture?

- (A) Elastic Compute Cloud (EC2)
- (B) Simple Storage Service (S3)
- (C) App Engine
- (D) Elastic Block Storage (EBS)

Q41. The lecture contrasts modern cloud programming paradigms like MapReduce with older ones. What older paradigm was mentioned for computation-intensive tasks on clusters or grids?

- (A) Object-Oriented Programming
- (B) MPI (Message Passing Interface) based high performance computing
- (C) Functional Programming
- (D) SQL for relational databases

Q42. What is the main benefit for startup companies to use public cloud resources, according to the lecture?

- (A) They can gain permanent ownership of servers at a low cost.
- (B) They can harness large computing resources without the upfront cost of buying their own machines.
- (C) They receive dedicated, in-person technical support from the cloud provider.

(D) They are guaranteed better network performance than established companies.

Q43. Microsoft's Bing search engine is used as an example of massive scale. How many servers were mentioned as running the Bing application?

- (A) 10,000
- (B) 40,000
- (C) 80,000
- (D) 150,000

Q44. Which programming paradigm, used by Facebook, is mentioned alongside Hadoop in the lecture?

- (A) Pig
- (B) Hive
- (C) Sawzall
- (D) Cassandra

Q45. When discussing datacenter cooling, the lecture mentions a metric 'annual water usage divided by the IT equipment energy'. What is a desirable value for this metric?

- (A) As high as possible
- (B) As low as possible
- (C) Exactly 1.0
- (D) The lecture does not state a preference

Q46. The lecture mentions that the definition of a 'computer' has shifted over time. What was the earliest concept in the sequence presented: 'The network is the computer', 'The data center is the computer', 'The cloud is the computer'?

- (A) The cloud is the computer
- (B) The data center is the computer
- (C) The network is the computer
- (D) All were proposed simultaneously

Q47. A research institute is analyzing a massive, static dataset of genomic information. The analysis requires a new parallel programming model. Which two features of cloud computing are most relevant to this project?

- (A) On-demand Access and Massive Scale
- (B) Data-intensive Nature and New Cloud Programming Paradigms
- (C) Massive Scale and Utility Computing
- (D) On-demand Access and Private Clouds

Q48. What is the primary characteristic of a 'service oriented architecture' (SOA) as framed within the lecture?

- (A) It is a precursor to peer-to-peer networks.
- (B) It is a hardware specification for datacenters.
- (C) It is a concept subsumed by Software as a Service (SaaS).
- (D) It is an alternative name for Infrastructure as a Service (IaaS).

Q49. What is the key advantage of a private cloud that is mentioned in the context of cost savings?

- (A) It requires no initial investment.
- (B) It allows a company to share computing power and storage resources across its internal servers.
- (C) It is managed by an external provider, reducing manpower costs.
- (D) It is always cheaper than a public cloud, regardless of use time.

Q50. The lecture describes how cloud computing realizes the vision of 'utility computing'. What is the core idea of this vision?

- (A) Every user owns and maintains their own server.
- (B) Computing is a paid service delivered by a provider, similar to electricity.
- (C) Software is a one-time purchase that runs forever.
- (D) Data is stored on local devices for maximum privacy.

Answer Key

Q1: Applications with variable workloads and large data sets

The transcript states at 00:52.410 that evolutionary changes were 'driven by the applications, with variable workload and large data sets.'

Q2: The virtualization of resources to create a flexible pool

At 15:42, the lecture explains that a cloud is built with 'physical or virtualized resources' and that 'virtualization of the resource will create pool of virtualized resources' which provides flexibility, distinguishing it from general distributed systems.

Q3: Mainframe timesharing systems

The lecture at 01:55.110 lists 'computing clusters, service oriented architecture, computational grids, peer to peer networks, internet cloud' as high-throughput systems. Timesharing is mentioned as an earlier historical phase (17:10), not as part of this specific group.

Q4: Saving significant time in server setup

The lecture states at 08:33 that a new server can be running in 'a fraction of minutes (3 minutes) compared to several weeks and months to purchase the server.' This highlights time savings as a major benefit.

Q5: On-demand Access

At 24:09, the lecture introduces 'On demand access' and immediately defines it as 'pay as you go, pay as you use'.

Q6: Amazon Web Service

The transcript at 03:54 identifies Amazon Web Service as the provider of EC2 and S3.

Q7: Platform as a Service (PaaS)

The lecture at 35:40 describes PaaS as providing access to infrastructure 'coupled with the programming platform.' Google App Engine is given as an example where users can develop applications within the framework.

Q8: Public clouds are open to any paying customer, while private clouds are for a single company's internal use.

The lecture defines a public cloud as providing 'services to any paying customer' (05:53) and a private cloud as being 'accessible internal to the companies' and 'managed internally by the company' (05:20-05:36).

Q9: David Patterson of UC Berkeley

At 11:54, the lecture states: 'in 2008 David Patterson of UC Berkeley has said that; the data center is the computer.'

Q10: To connect all the servers within a single rack

The transcript at 13:25 describes 'switches which are connecting these racks' and specifies that 'every rack will have a top of the rack switch, which is basically mentioned over here and they are connecting all the racks,' implying it connects the servers within that rack to the larger network.

Q11: The I/O performance of disk and network

At 38:31, the lecture states that in data-intensive computing, 'CPU utilization is no longer the important resource metric, but instead the IO input output that is the disk and also the network is important.'

Q12: Hadoop

The lecture mentions MapReduce and states at 25:56, 'and its open source version is called a new programming paradigm.' It later clarifies at 39:56 that Yahoo used 'Hadoop that is a open source version of the MapReduce.'

Q13: When the project duration is long (e.g., more than 6-12 months)

The analysis at 42:52 concludes that 'if the number of months is more than 6 months, then it is better to own the infrastructure that is go for a private cloud.'

Q14: Computing resources should be as easily accessible as electricity or water from a utility company.

The lecture at 21:40 explains Corbato's vision that a 'computing facility... like a power company or like a water company' should allow a user to 'plug-in into the computing utility and play' without knowing the underlying complexities.

Q15: Facebook

The transcript at 27:21 states that for the Facebook application, the server count grew to '180,000 servers are used to run one application that is Facebook'.

Q16: To provide compute services where users can run operating system images

The lecture at 06:49 describes Amazon EC2 (Elastic Compute cloud) as providing 'compute services' where a 'user can upload and run an arbitrary operating system images.'

Q17: Guaranteed Security

The four features listed at 23:10 are: Massive Scale, On-demand Access, Data-intensive Nature, and New Cloud Programming Paradigms. Security is mentioned as a risk for HaaS but not as one of the four defining features.

Q18: Power Usage Effectiveness (PUE)

The transcript at 31:43 describes this ratio as a measure of 'power utility' and this is the standard definition of PUE (Power Usage Effectiveness), which is contextually implied.

Q19: Peer-to-peer (P2P) systems

At 18:35, the lecture explicitly states that 'the precursor to the clouds are basically the peer to peer' systems.

Q20: Network bandwidth doubles each year.

The lecture at 20:34 states, 'Gilder's law indicates that the network bandwidth has doubled each year in the past.'

Q21: Data-intensive computing where computation moves to the data

The lecture at 37:24 describes data-intensive computing for applications with 'large data and data cannot be moved... but the computation has to move where ever the data is.'

Q22: Google Docs

The lecture at 36:34 provides 'Google's doc and MS office on demand' as examples of Software as a Service.

Q23: It presents security risks.

At 34:19, after describing HaaS, the lecture notes, 'But it is not a good idea because of the security risks.'

Q24: The servers used for submitting jobs and receiving client requests

At 14:03, the transcript states, 'Now, frontend will be there for submitting the jobs and receiving the client requests. So, often this can be treated as 3 tier architecture.'

Q25: Advances in virtualization

The lecture at 10:57 begins a section by stating, 'the advances in the virtualization makes it possible, the growth of internet clouds as a new computing paradigm.'

Q26: Based on the amount of space used, in GB per month

The lecture explains at 06:36 that for Amazon S3, users pay 'in terms of GB per month' for the storage space they use.

Q27: The operational cost is eliminated because customers only pay for what they use.

At 09:23, the lecture explains that online services reduce operational cost because it 'is not at all required. The only amount of money it has to be paid whatever as per the use.'

Q28: Cassandra

The transcript at 40:28 describes the key-value store 'Cassandra is also 2400 times faster than MySQL.'

Q29: Salesforce

At 04:47, the lecture lists 'Rightscale, Salesforce, EMC, Gigaspaces, 10gen...' as other cloud providers.

Q30: They are comprised of multiple datacenters in different locations connected by a network.

The lecture at 14:44 describes 'geographical distributed clouds' as having setups at 'more than one sites' which are 'multiple search datacenters which are connected together.'

Q31: The era of large datacenters like ENIAC in the 1940s-1960s.

At 18:15, the lecture states that current cloud systems have the 'same setup, that is called datacenters which were there in 1940s so and 1960s, so the same cycle is being repeated.'

Q32: Every 18 months

The lecture at 20:15 states, 'Moore's law indicates that the processor speeds doubles every 18 months.'

Q33: Developing software to be delivered as a service to millions of users.

At 11:09, the lecture highlights a 'dramatic difference between developing a software for millions to use as a service versus developing a software and distributing it to run on their own PCs.'

Q34: To connect all the 'top of the rack' switches together.

The lecture states at 14:14, 'Here there is a core switch which we will connect all the different top of the rack switches.'

Q35: Google

The transcript at 39:33 mentions, 'Google also uses the MapReduce in the form of indexing, for indexing it requires a chain of MapReduce of 24 MapReduce jobs.'

Q36: Rajkumar Buyya

The lecture at 12:11 says 'recently Rajkumar Buyya of Melbourne university simply said the cloud is the computer.'

Q37: Processing and analyzing petabytes of daily weblogs from a popular website.

The lecture defines the data-intensive feature by the large size of data (terabytes, petabytes) and gives 'daily logs' and 'weblogs' as examples at 25:15.

Q38: Hardware as a Service (HaaS)

At 34:56, the lecture states that 'infrastructure as a service has subsumed the hardware as a service,' making HaaS less used.

Q39: To be primarily used for storage purposes, often containing SSDs.

The lecture at 13:41 mentions 'storage backend nodes connected to the network' and clarifies at 13:53 that these nodes are 'primarily meant for the storage purposes. They are having the SSDs within it'.

Q40: App Engine

The lecture lists EC2, S3, and EBS as AWS services at 04:03. App Engine is identified as a service from Google at 04:33.

Q41: MPI (Message Passing Interface) based high performance computing

At 37:05, the lecture mentions that for earlier computation-intensive computing, 'MPI based high performance computing cluster or grids were formed.'

Q42: They can harness large computing resources without the upfront cost of buying their own machines.

The lecture states at 10:24 that 'startup companies they can harness large computing resources without buying their own machines.' This is linked to the economic decision of avoiding large capital expenses.

Q43: 80,000

At 28:01, the lecture states that '80,000 different servers were running one application which is called Bing.'

Q44: Hive

The transcript at 40:05 says, 'Facebook also uses the hadoop and the hive'.

Q45: As low as possible

At 31:33, after defining the water usage metric, the lecture states, 'if this particular parameter is low then it is good.'

Q46: The network is the computer

The lecture presents these chronologically, starting with John Gage's 1984 slogan 'the network is the computer' (11:46), followed by David Patterson's 2008 slogan, and finally Rajkumar Buyya's recent slogan.

Q47: Data-intensive Nature and New Cloud Programming Paradigms

The 'massive' dataset points to the 'Data-intensive Nature' feature. The need for a 'new parallel programming model' directly corresponds to the 'New Cloud Programming Paradigms' feature described in the lecture (24:44, 25:40).

Q48: It is a concept subsumed by Software as a Service (SaaS).

When discussing SaaS, the lecture at 36:19 states, 'And this will subsume the service oriented architectures.'

Q49: It allows a company to share computing power and storage resources across its internal servers.

The lecture states at 09:46 that a 'private cloud of the virtual servers inside the data centre has saved nearly crores of rupees annually because company can share; computing power and storage resources across the servers.'

Q50: Computing is a paid service delivered by a provider, similar to electricity.

The lecture at 22:56 defines utility computing as a 'business model in which customers receive computing services from a paid service provider,' using the analogy of a power company.